24 April 1969
SUPERSEDING
MIL-F-11698B (GL)
11 September 1964

MILITARY SPECIFICATION

FASTENER; QUICK RELEASE, ALUMINUM

1. SCOPE

* 1.1 This specification covers one type of cast aluminum, quick release fastener.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-A-591	-	Aluminum Alloy Die Castings.
QQ-A-596	-	Aluminum Alloy Permanent and Semipermanent Mold Castings.
QQ-A-601	-	Aluminum Alloys Sand Castings.
QQ-P-416	-	Plating, Cadmium (Electrodeposited).
QQ-S-698	-	Steel, Sheet and Strip, Low Carbon.
QQ-S-763	-	Steel Bars, Shapes and Forgings, Corrosion-Resisting.
QQ-W-321	-	Wire, Copper Alloy.
QQ-W-461	-	Wire, Steel, Carbon, (Round, Bare and Coated).
QQ-Z-325	-	Zinc Coating Electrodeposited, Requirements For.
PPP-B-636	-	Box, Fiberboard.
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of U. S. Military Property.
MIL-STD-147	-	Palletized and Containerized Unit Loads; 40 inch x 48 inch Pallets, Skids, Runners, or Pallet-Type Base.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T21:52Z
Check the source to verify that this is the current version before use.

DRAWING

Army Natick Laboratories

4-1-98 - Fastener; Quick Release, Aluminum.

(Miniature copy of Drawing 4-1-98, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL CLASSIFICATION BOARD

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W., Washington, D. C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Ill., 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), before production is commenced, a sample of the finished fastener shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

Source: https://assist.dla.mil -- Downloaded: 202610-17T21:52Z
Check the source to verify that this is the current version before use.

3.2 Materials.- Materials shall conform to the applicable specification and requirements specified herein.

3.2.1 Aluminum castings.- Aluminum castings shall conform to any one of the following methods.

(a) Sand castings. Sand castings shall conform to alloy no. 214 F. or 356-T6 of QQ-A-601.
(b) Die castings. Die castings shall conform to alloy no. 360 of QQ-A-591.
(c) Permanent mold castings. Permanent mold castings shall conform to alloy no. 356 of QQ-A-596.

3.2.2 Steel wire.- The steel wire shall conform to steel numbers 1010, 1015 or 1020, finish 1 of QQ-W-461.

* 3.2.3 Steel, sheet and strip.- The steel sheet shall conform to hot or cold rolled commercial quality of QQ-S-698. The steel strip shall conform to hot rolled commercial quality or cold rolled quarter hard no. 3 temper of QQ-S-698.

3.2.4 Corrosion-resisting steel.- The corrosion-resisting steel shall conform to class 302 of QQ-S-763.

* 3.2.5 Copper alloy wire.- The copper alloy wire shall conform to alloy no. 510 of QQ-W-321.

* 3.3 Construction.- The construction of the fastener shall conform to the design, shape and dimensions shown on Drawing 4-1-98 and as specified herein.

3.3.1 Body and latch assembly.- The body and latch shall be cast from aluminum as specified in 3.2.1.

3.3.2 Ball bearings.- The ball bearings shall be fabricated from corrosion-resisting steel as specified in 3.2.4.

3.3.3 Springs.- The springs shall be fabricated from copper alloy wire as specified in 3.2.5.

3.3.4 Pin.- The pin shall be fabricated from steel wire as specified in 3.2.2.

3.3.5 Plugs.- The plugs shall be fabricated from steel sheet or strip as specified in 3.2.3 or steel wire as specified in 3.2.2.

* 3.3.6 Link.- The link shall be cast from aluminum as specified in 3.2.1.

Source: https://assist.dla.mil -- Downloaded: 202610-17T21:52Z
Check the source to verify that this is the current version before use.

3.4 Finish.- The steel pin and plugs shall be given a coat of phosphate treated zinc plate conforming to type III, class 2 of QQ-Z-325 or a coat of cadmium plate conforming to type I, class 1 of QQ-P-416.

* 3.5 Marking for identification.- The fasteners shall be marked for identification in accordance with MIL-STD-130. The identification marks shall be incorporated in such a manner that the surfaces opposite the markings shall show no signs of penetration or unevenness or cause distortion in any part of the hardware.

* 3.6 Workmanship.- The finished fasteners shall be free of rough or sharp edges, pits, pores, blowholes, shrinkage or cracks.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for defects in table I and specified dimensions.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 End item inspection.- The lot shall consist of all complete quick release fasteners submitted for inspection at one time. The sample unit for this inspection shall be one complete quick release fastener.

* 4.3.2.1 Visual examination.- Examination shall be made of quick release fasteners defects listed in table I. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202640-17T21:52Z
Check the source to verify that this is the current version before use.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish (steel pins and plug)	Not cadmium or zinc plated		X
Construction and workmanship	Design or type not as specified	X	
	Blow holes, porosity, hard spots shrinkage or cracks	X	
	Sharp edges not removed	X	
	Less than 5 coils in spring	X	
Assembly	Any component missing or incorrectly assembled	X	
	Latch cannot be snaplocked in body with both retaining balls showing tension	X	
Identification marking	Missing, incomplete, illegible, or incorrect		X

4.3.2.2 Dimensional examination.- Examination shall be made of the quick release fasteners for defects in dimensions. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing, and marking are in compliance with scction 5 of this specification. Defects shall be scored as specified in table II. The sample unit for this examination shall be one shipping container fully prepared for delivery. The lot shall be the number of containers submitted for inspection at one time. The inspection level shall be S-2 with AQL of 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examination	Defect
Marking (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Material	Component missing, damaged, or otherwise defective.

Source: https://assist.dla.mil -- Downloaded: 202640-17T21:52Z
Check the source to verify that this is the currentversion before use.

TABLE II.- Examination of preparation for delivery (cont'd)

Examination	Defect
Workmanship	Inadequate application of components such as incomplete closure of container flaps, inadequate strapping or tape banding, stapling, bulging or distortion of containers.
Contents (exterior and interior container)	Number per container is more or less than required.

* 4.3.4 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing and strapping shall be in accordance with the appendix of the box specification.

* 4.3.5 Examination of palletization.- An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be scored as specified in table III. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5, expressed in terms of defects per hundred units.

TABLE III.- Examination of palletization

Examination	Defect
Finished dimension	Length, width or height exceeds specified maximum requirements.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limit.
Marking	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.

Source: https://assist.dla.mil -- Downloaded: 202640-17T21:52Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

* 5.1.1.1 Unit package.- The link shall be removed from the fastener and the latch shall be reset in the locked position. The link shall be positioned flat on the latch of the fastener with the larger end of the link at the hook end of the fastener body. Each fastener shall then be packaged in a close fitting clear polyethylene film bag of minimum 0.002 inch thickness. Closure shall be effected by heat sealing. Prior to or during the heat sealing closure operation, excess air within the bag shall be expelled.

* 5.1.1.2 Intermediate package.- Fifty quick release fasteners, unit packaged as specified in 5.1.1.1, shall be packaged in a snug fitting fiberboard box conforming to style RSC, type CF or SF, class domestic, variety and grade optional of PPP-B-636. Closure shall be in accordance with the appendix of PPP-B-636.

* 5.1.2 Level C.- Quick release fasteners shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

* 5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Quick release fasteners, packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing, and reinforcing with flat steel strapping or tape banding shall be in accordance with the appendix of the container specification.

* 5.2.2 Level B.- Quick release fasteners, packaged as specified in 5.1, shall be packed in a snug fitting fiberboard shipping container conforming to style RSC, type CF or SF, class domestic, variety optional, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the container specification.

5.2.2.1 When specified (see 6.2), the shipping container shall be a grade V3c or V3s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix of the container specification. The shipping container material may also be grade V4s of PPP-F-320.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T21:52Z
Check the source to verify that this is the current version before use.

* 5.2.3 Level C.- Quick release fasteners, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

* 5.3 Palletization.- Unless otherwise specified (see 6.2), quick release fasteners, packed as specified, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern intended to be used shall be submitted to and approved by the contracting officer.

5.4 Marking.- In addition to any special marking required by the contract or order, interior packages, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

* 6.1 Intended use.- The quick release fastener covered by this specification is used as a component of the single trace type man's harness.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) First article.- When preproduction sample is required (see 3.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant fiberboard boxes are required for level B shipment (see 5.2.2.1).
(e) When palletization is not required (see 5.2).

6.3 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:

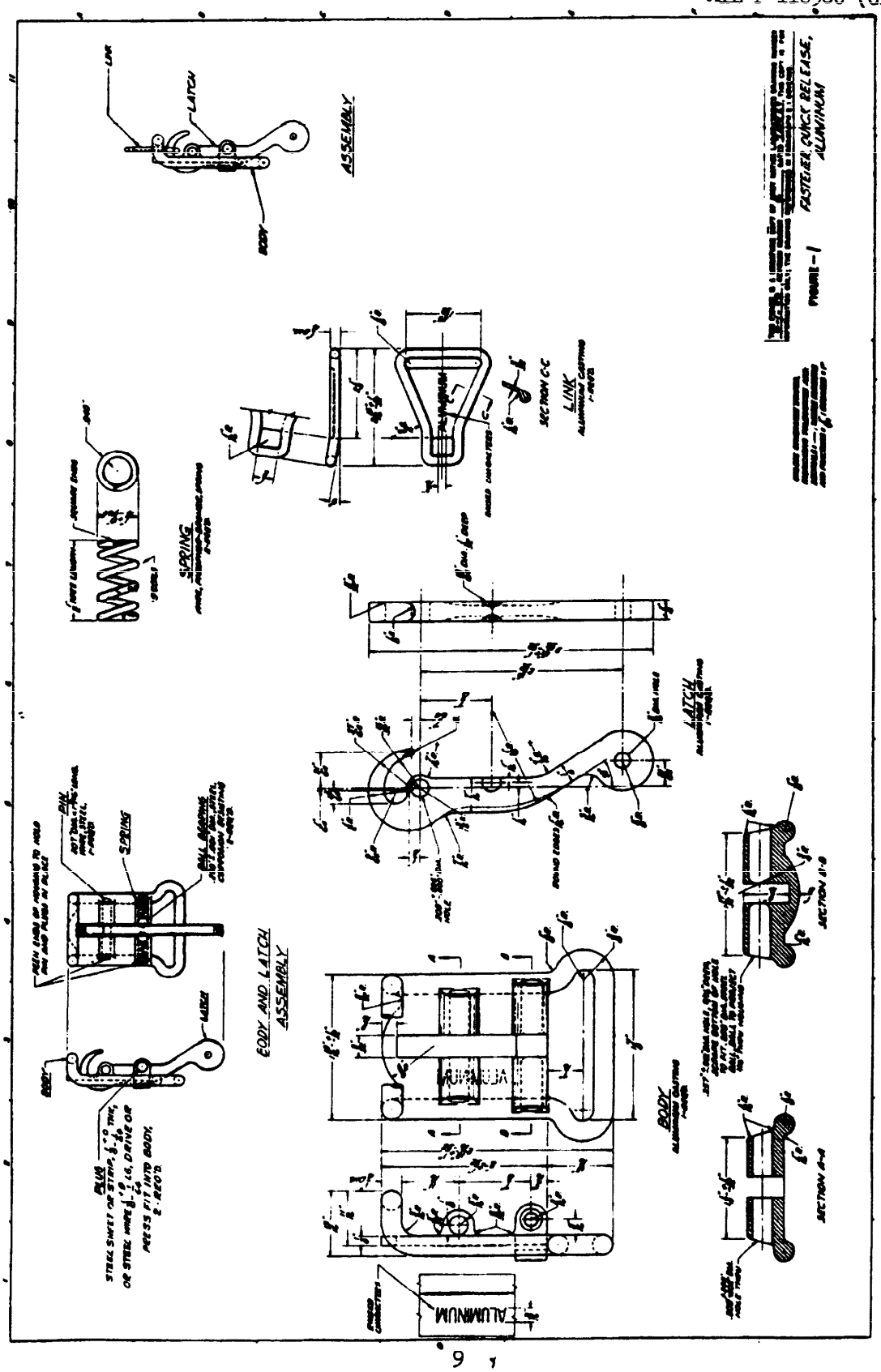
Army - GL

Preparing activity:

Army - GL

Project No. 5340-A001

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T21:52Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 202610-17T21:52Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Fastener; Quick Release, Aluminum MIL-F-11698C (GL)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD. FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T21:52Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY
U. S. Army Natick Laboratories
Natick, Massachusetts 01760

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Natick Laboratories
ATTN: AMXRES-EQS
Natick, Massachusetts 01760

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T21:52Z
Check the source to verify that this is the current version before use.